

Cancer as Progressive D-Collapse: A Tholonic Framework for Oncogenesis, Malignancy Grading, and Therapeutic Resistance

J. W. Milton

Clarity Coalition

14 June 2026

v1.0

Keywords: tholonic model; oncogenesis; tumor suppressor; D-collapse; balance score; malignancy grading; therapeutic resistance; hallmarks of cancer; N-D-C framework; quantitative oncology

Abstract

This paper applies the tholonic N-D-C (Negotiation-Definition-Contribution) framework to cancer biology, mapping the three structural roles onto cellular regulatory systems. N corresponds to cellular homeostasis (the stable, coherent state of a functioning cell), D corresponds to the constraint apparatus (tumor suppressor genes, cell cycle checkpoints, contact inhibition, and apoptosis signals), and C corresponds to the proliferative apparatus (growth factors, mitogenic signals, and metabolic outputs). Under this mapping, oncogenesis is a progressive D-collapse: each driver mutation removes or disables a D-primitive, incrementally lowering the balance score $B(D, C) = 2 \cdot \min(D, C) / (D + C) \cdot 100$. Metastasis is the limiting case in which C becomes entirely uncoupled from D, and N (homeostasis) dissolves.

The paper defines quantitative proxies for D and C that can be computed from tumor biopsy data: a Tumor Suppressor Gene Integrity (TSGI) score as the D-proxy, and the Ki-67 proliferative index as the primary C-proxy. It proposes that WHO histological malignancy grades correspond to four B-score bands separated at the tholonically significant thresholds of 80 and 61.8 ($1/\varphi \cdot 100$). Drug resistance is reframed as C-adaptation to an externally imposed D constraint, predicting that sequential therapy cycles should lower the floor B-score of a tumor over time.

This paper does not provide clinical trial data, does not propose treatment protocols, and does not claim that the tholonic framework supersedes existing oncological classification systems. It provides a structural organizing model and a set of falsifiable quantitative predictions that can be tested against existing genomic and histological datasets.

1 Introduction

Cancer kills approximately ten million people annually and remains the second leading cause of death globally [WHO23]. Despite decades of progress in molecular oncology, the field lacks a single unified structural framework that connects the molecular events of oncogenesis, the histological features of malignancy, and the mechanisms of therapeutic resistance under one consistent set of principles.

The tholonic N-D-C framework, introduced in paper 3 of this series [Mil26c] and grounded mathematically in paper 1 [Mil26a], describes how any coherent system maintains stability through the balance of two complementary structural roles: D (Definition, the constraint apparatus) and C (Contribution, the productive apparatus). The emergent state N is stable when $D \approx C$. When D erodes without a compensating reduction in C, the system becomes increasingly unstable and eventually incoherent.

Cancer fits this structural pattern precisely. The molecular oncology literature has independently described this pattern in detail: Hanahan and Weinberg’s “Hallmarks of Cancer” [Han00, Han11] enumerate the sequential acquisition of proliferative autonomy, resistance to growth suppressors, evasion of apoptosis, and invasion as the canonical trajectory of malignant transformation. Each hallmark is, in tholonic terms, either a D-erosion event or a C-amplification event. The tholonic framework does not add new biological facts; it provides a unifying structural language and a quantitative scoring model for the facts already established.

What this paper provides. A formal N-D-C mapping for normal cell biology; a D-collapse model of oncogenesis; quantitative D and C proxy definitions computable from biopsy data; a predicted correspondence between WHO malignancy grades and B-score bands; a structural account of drug resistance as C-adaptation; and four falsifiable predictions.

What this paper does not provide. Clinical treatment recommendations, primary experimental data, proof that B-scores outperform existing prognostic tools, or any claim that the tholonic framework replaces the molecular oncology literature it draws on.

Organization. §2 reviews the N-D-C framework. §3 maps the three roles onto cell biology. §4 develops the D-collapse model of oncogenesis. §5 defines quantitative proxies. §6 maps grades to B-score bands. §7 addresses drug resistance. §8 states falsifiable predictions. §9 discusses implications and limitations. §10 concludes.

2 The Tholonic N-D-C Framework

The tholonic model, developed in papers 1 through 3 of this series [Mil26a, Mil26b, Mil26c], holds that any self-sustaining system can be analyzed in terms of three structurally distinct roles.

N (Negotiation) is the emergent, coherent instantiation of the system at a given level: the stable state that exists as a consequence of D and C being approximately balanced. N is simultaneously the product of D and C at the current level and the source that differentiates

into D and C at the next level down. It is not directly measured; it is inferred from the persistence of the system's identity.

D (Definition) is the constraint apparatus: the rules, limits, boundaries, and specifications that define what the system is. D is internally focused. It governs structure, identity, and permissible states. High D means tightly bounded behavior.

C (Contribution) is the productive apparatus: the outputs, flows, connections, and applications that define what the system does. C is externally focused. It governs throughput, expression, and interaction with the environment. High C means high output and connectivity.

Balance condition. The scalar balance score is defined as:

$$B(D, C) = \frac{2 \cdot \min(D, C)}{D + C} \cdot 100$$

A score of 100 means perfect balance. Scores above 80 indicate coherent operation. Scores between 61.8 and 80 indicate early imbalance. Scores below 61.8 (the $1/\varphi$ threshold, where φ is the golden ratio) indicate structural instability. Scores below 38.2 indicate near-collapse. These thresholds are derived from the mathematical structure of the framework (paper 2, [Mil26b]) and are not fitted to the oncology data.

Recursive structure. The full tholonic model is hierarchical: each N at one level becomes the parent that differentiates into D and C at the level below. For this paper, we operate at a single level: the individual cell as the system under analysis, with cellular homeostasis as N.

3 N-D-C Mapping for Normal Cell Biology

A normal, healthy cell in a differentiated tissue maintains a stable identity and controlled behavior. Under the tholonic mapping:

N: Cellular Homeostasis. The stable, coherent state of a normally functioning cell: controlled proliferation rate, intact differentiation identity, appropriate apoptotic responsiveness, and respect for tissue boundaries. N is not a single molecule or pathway; it is the emergent functional state that results from D and C being in approximate balance.

D: The Tumor Suppressor and Checkpoint Apparatus. The D role in cell biology is occupied by the collection of mechanisms that define and constrain cellular behavior:

- Tumor suppressor genes (TP53, RB1, BRCA1, BRCA2, PTEN, APC, VHL, CDH1, SMAD4)
- Cell cycle checkpoints: the G1/S checkpoint (TP53/RB axis), the G2/M checkpoint, and the spindle assembly checkpoint
- Contact inhibition: the suppression of proliferation upon cell-to-cell contact
- Apoptotic machinery: BAX, APAF1, caspase cascades, anoikis
- DNA damage response: ATM, ATR, CHEK1, CHEK2

N-D-C Role Mapping: Normal Cell Biology

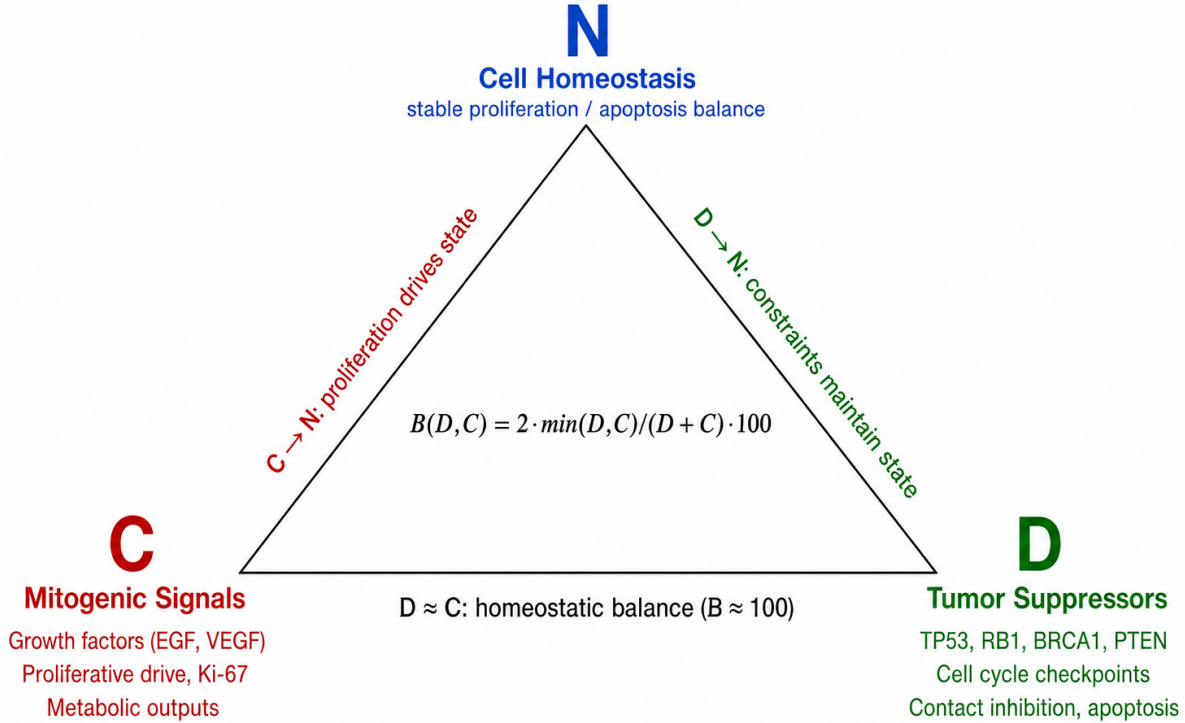


Figure 1: The N-D-C role mapping for normal cell biology. N (blue, top vertex) is cellular homeostasis; C (red, lower-left) is the mitogenic and proliferative apparatus; D (green, lower-right) is the tumor suppressor and checkpoint apparatus. The balance condition $B(D, C) \approx 100$ holds in healthy tissue.

Each of these is a D-primitive: a discrete constraint that, when functional, prevents the cell from transitioning to an uncontrolled proliferative state.

C: The Mitogenic and Proliferative Apparatus. The C role is occupied by the mechanisms that drive productive cellular output:

- Growth factor signaling: EGF/EGFR, VEGF, FGF, IGF-1
- Mitogenic signal transduction: RAS/MAPK, PI3K/AKT/mTOR
- Metabolic outputs: glucose uptake (GLUT1), glutamine metabolism, lipid synthesis
- Proliferative rate, quantified by Ki-67 expression and mitotic index
- Angiogenic signals: VEGF secretion, HIF-1 α activation under hypoxia

In a normal cell, these drives are constrained by D, and the resulting balance sustains N (homeostasis). The balance score is near 100.

4 Oncogenesis as Progressive D-Collapse

4.1 Driver Mutations as D-Primitive Loss Events

The molecular oncology literature distinguishes driver mutations (those that confer selective growth advantage) from passenger mutations (background noise) [Vog13]. Under the tholonic mapping, driver mutations are, almost exclusively, D-primitive loss events. The most common driver mutations in human cancers confirm this:

- **TP53 loss** (mutated in $>50\%$ of all cancers [Lev20]): removes the G1/S checkpoint anchor and the primary apoptotic trigger. A single D-primitive, absent in more than half of all human tumors.
- **RB1 loss**: removes the restriction point governor of the cell cycle. D-primitive loss.
- **PTEN loss**: removes the primary brake on PI3K/AKT proliferative signaling. D-primitive loss that simultaneously increases C (AKT/mTOR activation).
- **APC loss** (in $>80\%$ of colorectal cancers [Fea01]): removes the WNT pathway brake. D-loss with secondary C-amplification through β -catenin.
- **CDKN2A loss**: removes p16/p14ARF, disabling both RB1 and TP53 pathways simultaneously. A single mutation that erases two D-primitives.
- **SMAD4 loss**: removes TGF- β growth-inhibitory signaling. D-primitive loss.

Activating oncogenic mutations (KRAS G12D, PIK3CA H1047R, BRAF V600E) operate differently: they constitutively amplify C signaling without requiring growth factor input. These are C-amplification events rather than D-erosion events but produce the same structural consequence: $C > D$ and B-score falls.

The Knudson “two-hit hypothesis” [Knu71], which holds that both alleles of a tumor suppressor must be inactivated to remove its protective effect, is in tholonic terms the statement that a D-primitive is not lost until both copies are disabled. A heterozygous carrier has D partially degraded at that locus; full loss requires the second hit.

4.2 The Hanahan-Weinberg Hallmarks as a D-Collapse Sequence

Hanahan and Weinberg [Han00, Han11] describe the acquisition of cancer hallmarks as a stepwise process. In tholonic terms, each hallmark corresponds to a specific D-erosion or C-amplification event:

Hallmark	Tholonic interpretation
Sustaining proliferative signaling	C-amplification (constitutive RAS, EGFR, HER2 activation)
Evading growth suppressors	D-erosion (RB1, CDKN2A, TGF- β pathway loss)
Resisting cell death	D-erosion (TP53, BCL2 family dysregulation)
Enabling replicative immortality	D-erosion (telomere maintenance via TERT reactivation)

Hallmark	Tholonic interpretation
Inducing angiogenesis	C-amplification (VEGF, HIF-1 α upregulation)
Activating invasion and metastasis	C fully uncoupled from D (EMT, MMP upregulation, anoikis resistance)
Reprogramming energy metabolism	C-amplification (Warburg effect, glutamine dependency)
Evading immune destruction	D-erosion at the tissue-immune interface (PD-L1 upregulation, MHC-I loss)

No hallmark in the original or updated list represents a C-erosion event without a compensating D-amplification. Cancer never “slows down from excess D.” It always moves in the direction of D-erosion relative to C.

4.3 Metastasis as Complete C-D Uncoupling

Metastasis is the clinical inflection point at which the prognosis for most cancers changes dramatically. Under the tholonic model, metastasis is the structural event corresponding to C becoming completely uncoupled from D.

In a primary tumor, some D-primitives remain functional: the tumor is constrained spatially (basement membrane integrity), the cells retain partial differentiation identity (E-cadherin expression), and the D of the surrounding stroma provides some residual constraint. In metastasis:

- Epithelial-mesenchymal transition (EMT) dissolves the cell’s D-defined identity (E-cadherin loss, vimentin gain)
- Matrix metalloproteinase (MMP) secretion degrades the physical D-substrate (basement membrane)
- Anoikis resistance removes the apoptotic D-signal triggered by loss of matrix attachment
- The circulating tumor cell (CTC) phase is a period in which the cell exists with near-zero D: no tissue identity, no positional constraint, no growth factor-dependent signaling requirement

The resulting state (N dissolves; the cell is no longer “a liver cell” or “a lung cell” but generalized malignant cell) is precisely the tholonic prediction for the consequence of $D \rightarrow 0$ while C remains high.

5 Quantitative Proxies for D and C

5.1 The Tumor Suppressor Gene Integrity Score (TSGI)

To compute a D-proxy from biopsy or genomic data, we define the Tumor Suppressor Gene Integrity Score (TSGI) as:

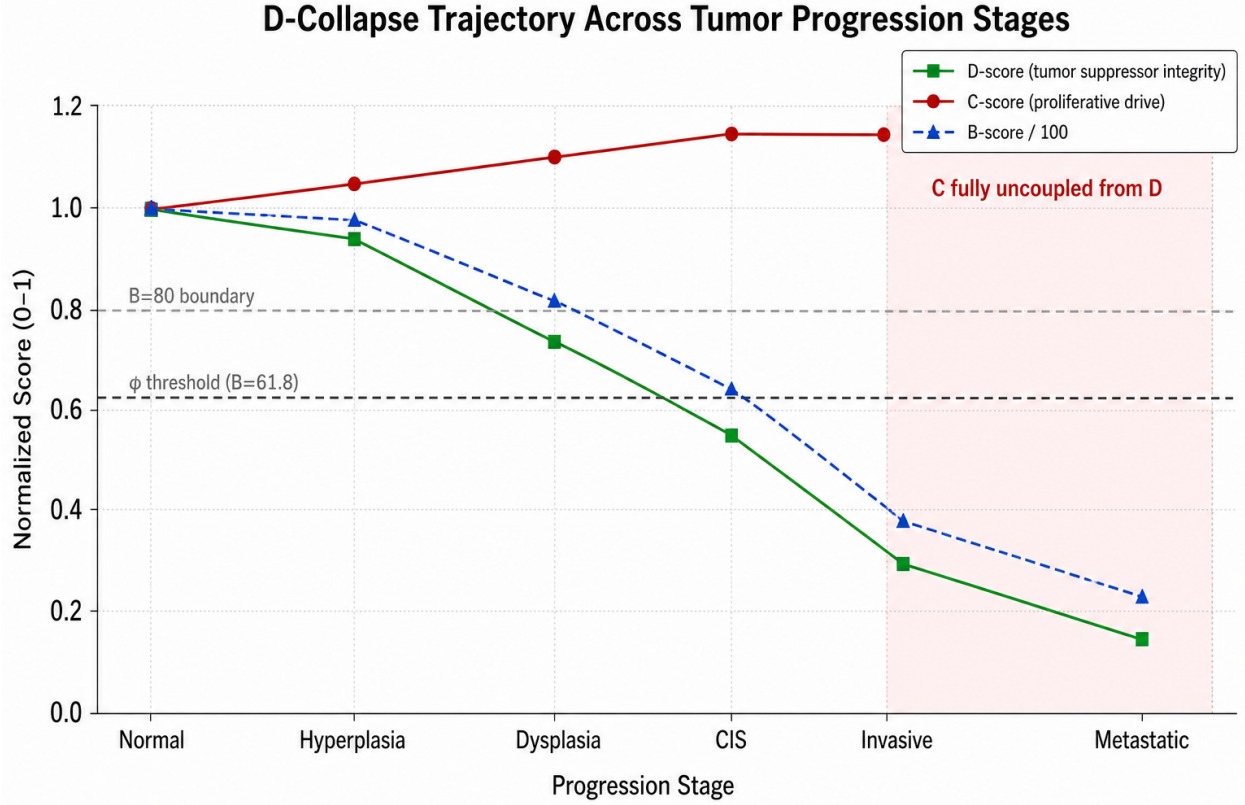


Figure 2: D-collapse trajectory across tumor progression stages. The D-score (green, tumor suppressor integrity) decreases monotonically while C-score (red, proliferative drive) rises slightly. The B-score (blue dashed) falls through the φ threshold ($B=61.8$) at the carcinoma in situ (CIS) stage, marking the boundary beyond which the structural instability is severe.

$$\text{TSGI} = \frac{\sum_{i=1}^k w_i \cdot s_i}{\sum_{i=1}^k w_i}$$

where k is the number of D-primitive genes in the panel, w_i is the weight assigned to gene i (reflecting its functional importance and the frequency with which its loss drives cancer), and $s_i \in \{0, 0.5, 1\}$ is the integrity score for that gene ($s_i = 1$ for intact, $s_i = 0.5$ for heterozygous loss, $s_i = 0$ for biallelic loss or silencing).

A minimal panel sufficient to capture the major D-primitives across common cancers includes: TP53, RB1, PTEN, APC, BRCA1, BRCA2, CDKN2A, VHL, SMAD4, and CDH1. Weights can be set uniformly ($w_i = 1$) for initial studies; differential weighting based on cancer-type-specific driver frequencies can be applied for tissue-specific analyses.

The TSGI ranges from 0 (all D-primitives lost) to 1 (all intact), and serves as the normalized D-input to the balance formula.

5.2 The Proliferative Index as C-Proxy

The C-proxy is defined as the normalized proliferative index PI:

$$\text{PI} = \frac{\text{Ki-67\%}}{100}$$

Ki-67 is a nuclear protein expressed in all phases of the cell cycle except G0. Its immunohistochemical quantification as a percentage of tumor cells is a standard clinical measurement with established prognostic value across multiple cancer types [Sch11]. Values above 20% are typically associated with high-grade tumors.

For a fully quiescent normal cell, $\text{PI} \approx 0.05$. For a Grade IV glioblastoma, PI may reach 0.40 to 0.90. Normalizing against the baseline expected in the tissue type of origin allows cross-tissue comparisons.

5.3 Computing the B-Score

Given TSGI and PI computed from a tumor sample, the balance score is:

$$B = \frac{2 \cdot \min(\text{TSGI}, 1 - \text{PI}_{\text{norm}})}{(\text{TSGI}) + (1 - \text{PI}_{\text{norm}})} \cdot 100$$

where PI_{norm} is the PI normalized to a 0-1 scale relative to the expected maximum for the tissue type, so that $1 - \text{PI}_{\text{norm}}$ represents the residual proliferative constraint. The formula treats D (TSGI) and the C-inverse ($1 - \text{PI}_{\text{norm}}$) symmetrically, consistent with the framework’s definition.

An alternative and more robust C-proxy is the ratio of the tumor’s mitotic count to the tissue-specific baseline, combined with tumor mutational burden (TMB, in mutations per megabase), where TMB serves as a measure of accumulated C-runaway (genomic instability driven by C-amplified replication with inadequate D-mediated repair).

5.4 Worked Example: Colorectal Adenocarcinoma

A colorectal adenocarcinoma with the following genomic profile:

- APC biallelic loss ($s = 0, w = 1$)
- TP53 biallelic loss ($s = 0, w = 1$)
- KRAS G12D activating mutation (C-amplifier, not D-loss)
- SMAD4 intact ($s = 1, w = 1$)
- All other panel genes intact

$\text{TSGI} = (0 + 0 + 1 + 1 + 1 + 1 + 1 + 1 + 1)/9 = 0.78$ (for a 9-gene panel excluding KRAS which is a C-primitive).

Ki-67 measured at 35% in a colonic tissue baseline of 5%: $\text{PI}_{\text{norm}} = (35 - 5)/(90 - 5) = 0.35$.

$$1 - \text{PI}_{\text{norm}} = 0.65.$$

$$B = 2 \cdot \min(0.78, 0.65)/(0.78 + 0.65) \cdot 100 = 2 \cdot 0.65/1.43 \cdot 100 \approx 91.$$

This intermediate result reflects a tumor with meaningful D-loss but moderate C: consistent with a WHO Grade II or early Grade III colorectal tumor with intact mismatch repair. Adding SMAD4 loss ($s = 0$, TSGI drops to 0.67) and increasing Ki-67 to 60% ($\text{PI}_{\text{norm}} = 0.65$, $1 - \text{PI}_{\text{norm}} = 0.35$) yields $B = 2 \cdot 0.35/(0.67 + 0.35) \cdot 100 \approx 69$, placing it below the Grade III boundary.

6 Predicted Correspondence with WHO Malignancy Grades

The WHO Classification of Tumours [WHO22] uses a four-tier grading system (Grade I through IV) based on histological features including mitotic activity, nuclear pleomorphism, necrosis, and vascular proliferation. These features are, in tholonic terms, observable consequences of D-collapse at different stages.

The tholonic model predicts the following correspondence:

WHO Grade	Description	Predicted B-score range	Tholonic interpretation
Grade I	Well-differentiated, slow-growing, minimal mitoses	$B \geq 80$	Few D-primitives lost; $D \approx C$
Grade II	Moderately differentiated, low-intermediate mitoses	$61.8 \leq B < 80$	1 to 2 D-primitives lost; early imbalance

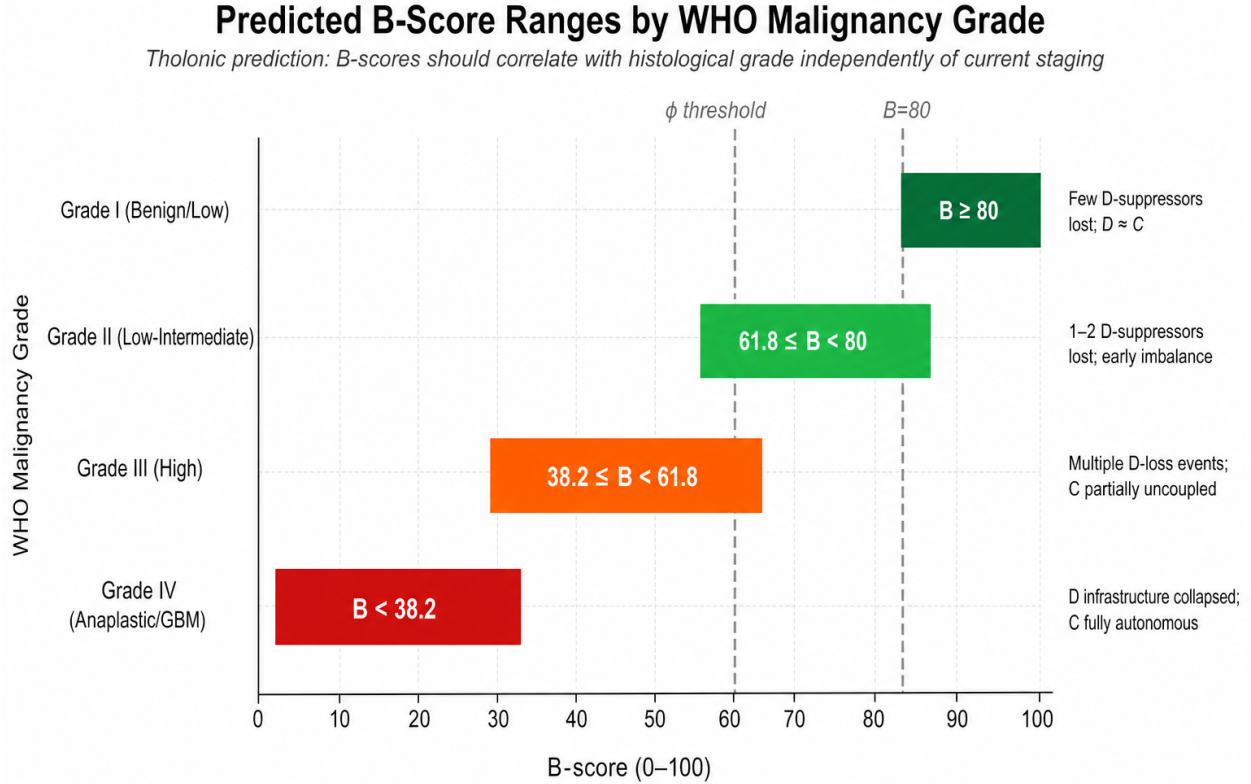


Figure 3: Predicted B-score ranges by WHO malignancy grade. Grade boundaries correspond to the tholonic thresholds of $B=80$ and $B=61.8$ ($1/\varphi \cdot 100$). The grade descriptions on the right reflect the biological consequences of progressively deeper D-collapse.

WHO Grade	Description	Predicted B-score range	Tholonic interpretation
Grade III	Poorly differentiated, high mitoses, pleomorphism	$38.2 \leq B < 61.8$	Multiple D-loss events; C partially uncoupled
Grade IV	Anaplastic or undifferentiated; necrosis, high mitoses	$B < 38.2$	D infrastructure collapsed; C fully autonomous

The grade boundaries at $B = 80$ and $B = 61.8$ are not fitted to the grade definitions. They are the same thresholds used in paper 2 [Mil26b] for supply chain transparency classification and follow from the mathematical properties of the balance formula and the golden ratio. The fact that these thresholds may align with empirically established tumor grade boundaries is a prediction, not a design feature.

7 Drug Resistance as C-Adaptation to Imposed D

7.1 Chemotherapy and Targeted Therapy as External D Constraints

Cytotoxic chemotherapy acts primarily as a C-suppressor: it interferes with DNA replication (C-output) or mitotic machinery (C-amplification machinery), thereby reducing C without directly restoring D. Targeted therapies (kinase inhibitors, receptor antagonists) act more specifically: they block constitutively active C-amplifiers (EGFR inhibitors, BRAF inhibitors, BCR-ABL inhibitors) without restoring any lost D-primitive.

In tholonic terms, a drug that suppresses C creates a transient B-score improvement (C falls toward D). This is not a genuine D-restoration; it is an externally imposed D-like constraint that the tumor's cellular machinery is already under selective pressure to circumvent.

7.2 Resistance Mechanisms as C-Routing

The well-documented mechanisms of therapeutic resistance map directly onto C-adaptation to a new D constraint:

- **Bypass signaling** (e.g., MET amplification following EGFR inhibition): C routes around the blocked node to a parallel amplification pathway. The new D (inhibitor) is bypassed, and C is re-established through an alternative channel.
- **Efflux pump upregulation** (P-glycoprotein/ABCB1 in multi-drug resistance): C produces a mechanism that actively removes the D-imposing agent from the cell.
- **Target mutation** (T790M EGFR, T315I BCR-ABL): the D-constraint (the drug's binding site) is structurally altered so C can no longer be blocked by it.
- **Lineage switching** (e.g., NSCLC to SCLC transformation under ALK inhibition): the cell changes its D-defined identity to one where the imposed D is no longer relevant.

Each resistance mechanism is a C-adaptation to a specific D-imposition. The resulting resistant clone is a new child N with a lower floor B-score: it has lost whatever D-primitives it originally possessed and has added C-adaptations that make future D-impositions harder to sustain.

7.3 Immunotherapy as D-Restoration

Immune checkpoint inhibitors (anti-PD-1, anti-PD-L1, anti-CTLA-4) operate differently from other therapies. Rather than imposing an external D or suppressing C, they restore a D-primitive that the tumor has specifically eroded: the immune system's ability to recognize and constrain malignant cells. PD-L1 upregulation in tumors is a D-erosion event at the tumor-immune interface (it disables the immune system's constraint on the tumor). Anti-PD-1 therapy restores this D-primitive.

This structural difference predicts different resistance mechanisms. Resistance to checkpoint inhibitors should primarily manifest as further D-erosion at the immune interface

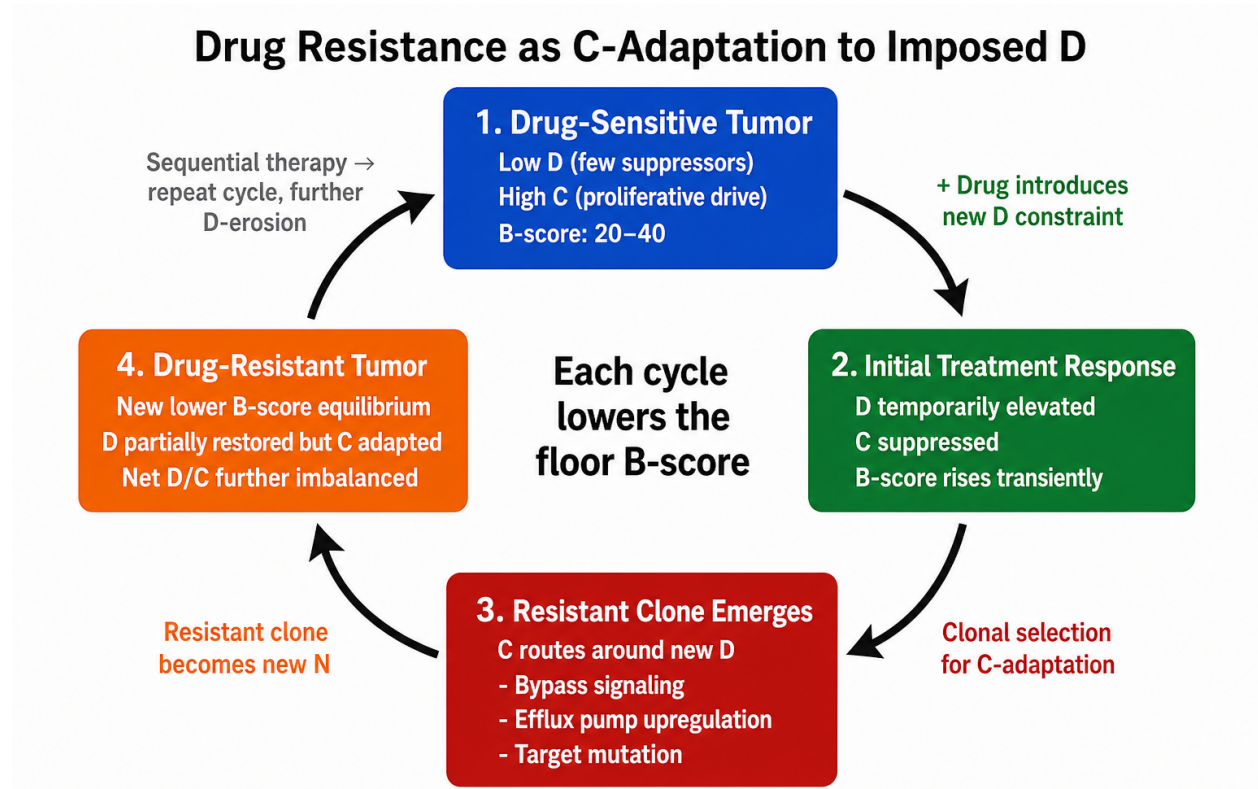


Figure 4: Drug resistance as C-adaptation to imposed D. Each cycle of therapy introduces a new D constraint; the tumor evolves C to route around it; the resistant clone becomes the new N; sequential cycles progressively lower the floor B-score.

(MHC-I loss, beta-2-microglobulin mutation, downstream IFN- γ signaling loss), not as C-routing, because the therapy is itself D-restorative rather than C-suppressive.

8 Falsifiable Predictions

The following predictions follow from the tholonic mapping and can be tested against existing datasets (TCGA, ICGC, METABRIC, SEER-linked genomic cohorts) without new data collection.

Prediction 1: B-score predicts progression-free survival independently of WHO grade. Tumors with TSGI and Ki-67 values placing them in a lower B-score band should have shorter progression-free survival and higher recurrence rates, independent of their WHO grade assignment. The prediction is falsified if B-score adds no prognostic information beyond grade in a Cox proportional hazards model with grade as covariate.

Prediction 2: Pre-treatment B-score predicts time to resistance. Tumors with lower B-scores at the time of first-line therapy initiation should develop resistance to that therapy more rapidly than tumors with higher B-scores (after controlling for tumor type and

drug class). The structural basis: a tumor with more D-primitives already lost has a larger C-adaptation repertoire available. The prediction is falsified if pre-treatment B-score has no association with time to resistance in a matched cohort analysis.

Prediction 3: Immunotherapy responders have higher pre-treatment D-proxy scores. Patients who respond durably to checkpoint inhibitor therapy should have higher TSGI scores (more intact D-primitives) and lower TMB (less C-runaway) at baseline than non-responders, after controlling for tumor mutational burden as a predictor of neoantigen load. The structural basis: a tumor with more residual D is more amenable to D-restoration therapy. The prediction is falsified if TSGI has no association with durable response in a checkpoint inhibitor cohort.

Prediction 4: B-score trajectories are monotonically decreasing in progressive disease. In patients with sequential tumor biopsies (paired primary/metastatic, or longitudinal liquid biopsies), the B-score computed from each sample should be lower in later samples than earlier ones in patients with progressive disease, and stable or increasing in patients with durable response. The prediction is falsified if B-score trajectories are non-monotonic at rates comparable to control distributions.

9 Discussion

9.1 Relationship to Existing Oncological Frameworks

The tholonic D-collapse model is not a replacement for existing oncological frameworks. The Hallmarks of Cancer [Han00, Han11] provide the biological detail; the clonal evolution model [Now06] explains the dynamics; the cancer stem cell hypothesis addresses hierarchical tumor organization. The tholonic model provides a unifying structural language that can be applied across these frameworks without contradiction.

The closest existing quantitative analogue is the concept of “cancer evolutionary fitness” as developed in evolutionary oncology [Gal20], which treats tumor clones as units of selection under environmental constraints. In tholonic terms, D-primitives function as the selection environment (the constraints), and C-adaptations are the fitness-improving mutations. The balance score B tracks the degree to which selection pressure (D) can still shape the evolutionary trajectory (C).

9.2 The Role of Tumor Heterogeneity

Intratumoral heterogeneity [McGr13] complicates the application of a scalar B-score to a tumor. Different subclones within a single tumor may have different D-primitive profiles and therefore different local B-scores. The framework accommodates this: the tumor-level B-score should be computed as a distribution rather than a single value, with the minimum B-score subclone being the most clinically relevant (it represents the most advanced D-collapse and the greatest resistance potential).

Liquid biopsy circulating tumor DNA (ctDNA) panels, which can track the allele frequencies of driver mutations over time, provide the data infrastructure needed to compute B-score distributions and their temporal trajectories without serial biopsies.

9.3 Limitations

The TSGI as defined in §5.1 weights tumor suppressor genes uniformly by default. Tissue-specific weighting, which would reflect the differential importance of specific D-primitives in different cancer types, requires validation against cancer-specific datasets. Uniform weighting is a conservative starting point.

The Ki-67 proliferative index, while widely measured, has known inter-laboratory variability [Dow11]. More robust C-proxies (TMB, mitotic count, standardized Ki-67 protocols) should be incorporated in validation studies.

The four falsifiable predictions in §8 are stated at the level of population-level correlations. Individual patient B-scores will be noisy. The framework claims a structural relationship, not a deterministic rule.

10 Conclusions

Cancer is not a random accumulation of mutations. It is a structured progression from D-C balance toward D-collapse, driven by discrete D-primitive loss events (tumor suppressor gene inactivation) and C-amplification events (oncogene activation), culminating in a state where C operates without any meaningful D constraint.

The tholonic N-D-C framework makes this structure explicit and quantitative. The TSGI and Ki-67 proxies defined here allow the balance score $B(D, C)$ to be computed from standard clinical biopsy data. The predicted correspondence with WHO grades and the four falsifiable predictions provide concrete testable claims. The structural reframing of drug resistance as C-adaptation to externally imposed D accounts for the universality of resistance across drug classes and suggests why D-restorative therapies (checkpoint inhibitors) have a different resistance profile from C-suppressive therapies (cytotoxics, kinase inhibitors).

If the predictions in §8 are borne out by retrospective analysis of existing genomic cohorts, the tholonic balance score could serve as a complementary prognostic and predictive biomarker alongside, but not replacing, established grading and staging systems.

11 References

[Dow11] Dowsett, M., et al. Assessment of Ki-67 in breast cancer: recommendations from the International Ki-67 in Breast Cancer Working Group. *Journal of the National Cancer Institute* 103(22), 2011.

Gene	Primary D-primitive function	Cancer types where loss is driver
SMAD4	TGF- β growth-inhibitory signaling	Colorectal, pancreatic
CDH1	E-cadherin: contact inhibition, EMT brake	Gastric (diffuse), lobular breast, colorectal

TSGI computation steps:

1. For each gene i , determine status from genomic data: intact (score 1), heterozygous loss (score 0.5), biallelic loss or promoter methylation silencing (score 0).
2. Apply weights: $w_i = 1$ for uniform panel.
3. $\text{TSGI} = \sum w_i s_i / \sum w_i$.
4. Compute PI_{norm} from Ki-67 % in the sample, normalized to tissue baseline.
5. $B = 2 \cdot \min(\text{TSGI}, 1 - \text{PI}_{\text{norm}}) / (\text{TSGI} + 1 - \text{PI}_{\text{norm}}) \cdot 100$.

For tumors lacking Ki-67 measurements, TMB (mutations per megabase, normalized to the tissue-type median) may substitute as the C-proxy after validation in the target tumor type.